Enrolment No.:

AISSMS
INSTITUTE OF INFORMATION TECHNOLOGY
ADDING VALUE TO ENGINEERING

Total No. of Questions: 8] **[Total No. of Printed Pages: 2**

BTECH_ET_SEM5 / ET Fifth Semester
ETPCC502 / Electromagnetic Theory
(May - 2026, 2025/2024 Pattern)

Time: 2:30 Hrs. **Maximum Marks: 60**

Instructions: -

(i) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
(ii) Neat diagrams must be drawn wherever necessary.
(iii) Figures to the right indicate full marks.
(iv) Assume suitable data, if necessary.
(v) Only Scientific calculators are allowed.

Q1. A) State and explain the Biot- savart law with mathematical expression. **[7] CO3 [UNDERSTANDING]**

B) A 6m long straight filamentary conductor carrying a current of 4A is placed along positive z axis its one end at origin.Find out the magnetic field intensity at point P (0,3,3)m. **[8] CO3 [UNDERSTANDING]**

(OR)

Q2. A) Using Biot Savart law, find H due to straight conductor of finite length carrying current of I amperes. **[7] CO3 [UNDERSTANDING]**

B) Verify stokes theorem for the field H = 6xy a_x -3y^2 a_y and rectangular path around the region, 2≤ x ≤ 5, -1≤y≤1, z=0.Let the positive direction of ds be a_z . **[8] CO3 [UNDERSTANDING]**

Q3. A) Explain the effect of field on conductor and dielectric. **[7] CO4 [UNDERSTANDING]**

B) Given that **[8] CO4 [UNDERSTANDING]**
E1 = 2ax - 3ay + 5az V/m at the charge free dielectric interface where εr1 = 2 and εr2= 5.find E2 & D2 .

(OR)

Q4. A) Derive the boundary condition for the interface between two magnetic media of different permeability. **[7] CO4 [UNDERSTANDING]**

B) Let the permittivity be $5\mu H/m$ in region A where x < 0 and $20\mu H/m$ in region B where x > 0 If there is surface current density K = 150 $\bar{a}_y$ - 200 $\bar{a}_z$ A/m . at x=0 and if H_A = 300 $\bar{a}_x$ - 400 $\bar{a}_y$ + 500 $\bar{a}_z$ A/m . Find : (i) $|H_{tA}|$ (ii) $|H_{nA}|$ (iii) $|H_{tB}|$ (iv) $|H_{nB}|$ **[8] CO4 [UNDERSTANDING]**

Q5. A) State faraday's and Lenz's law also explain the different ways of e.m.f generartion. **[7] CO5 [UNDERSTANDING]**

B) If D = 10x $\bar{a}_x$ – 4y $\bar{a}_y$ + k $\bar{a}_z$ μ C/m^2 and B = 2 $\bar{a}_y$ mT.Find the value of k to satisfy the Maxwell equations for the region σ = 0 and ρv =0. **[8] CO5 [UNDERSTANDING]**

(OR)

Q6. A) Write Maxwell's equation in point form and integral form for time varying electric and magnetic fields. **[7] CO5 [UNDERSTANDING]**

B) Given H = Hm $e^{j(\omega t + \beta z)}$ $\bar{a}_x$ A/m in free space.Find E **[8] CO5 [UNDERSTANDING]**

Q7. A) Derive general wave equation for Uniform plane wave for free space.E and H fields using Maxwell's equation. **[7] CO6 [APPLYING]**

B) E and H wave travelling in free space, are normally incident on the surface with a perfect dielectric with εr = 3.Compute the magnitude of incident, reflected and transmitted E and H wave at interface. **[8] CO6 [APPLYING]**

(OR)

Q8. A) Define phase velocity and group velocity also explain the same. **[7] CO6 [APPLYING]**

B) An E field in free space is given as E = 800 cos(10^8 t- βy) $\bar{a}_z$ V/m. Find **[8] CO6 [APPLYING]**

a) β b) λ c) H at P (0.1,1.5,0.4) at t= 8 nsec.

*****